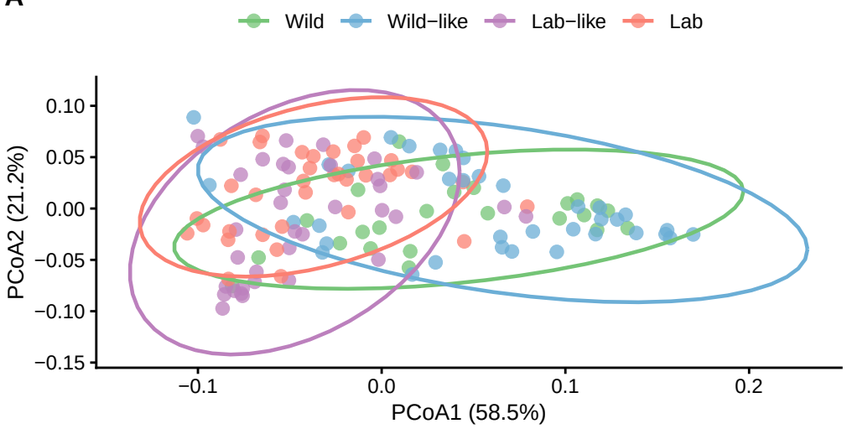
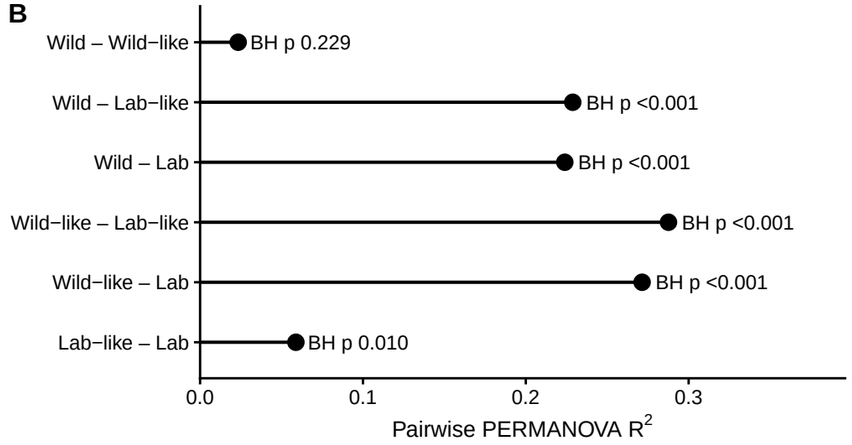
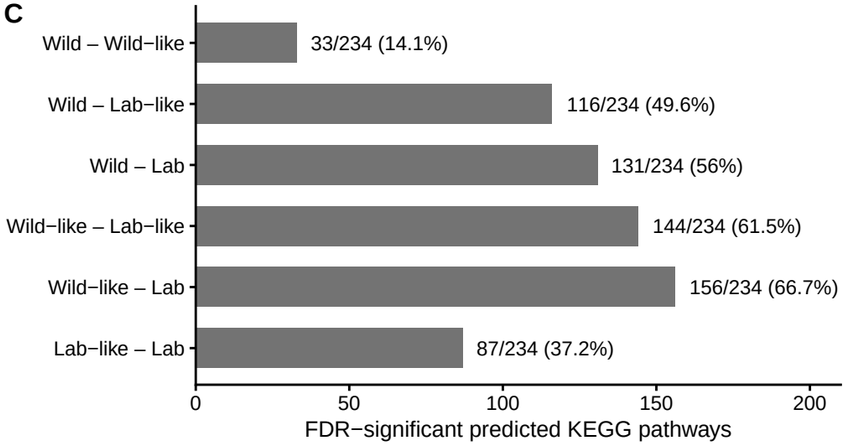
A


Wild
Wild−like
Lab−like
Lab
0.10
0.05
0.00
−0.05
−0.10
−0.15
PCoA2 (21.2%)
−0.1
0.0
0.1
0.2
PCoA1 (58.5%)

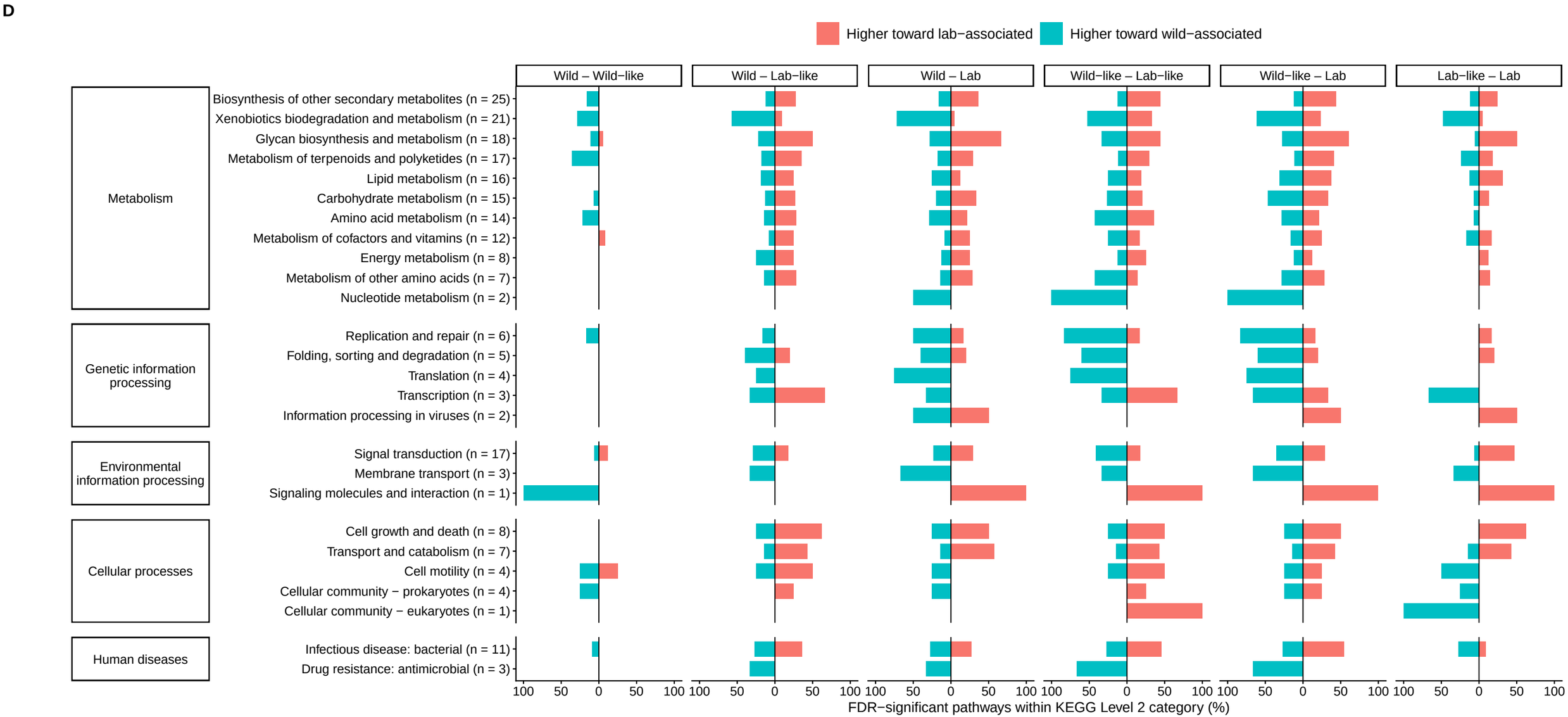
B


Wild – Wild−like
BH p 0.229
Wild – Lab−like
BH p <0.001
Wild – Lab
BH p <0.001
Wild−like – Lab−like
BH p <0.001
Wild−like – Lab
BH p <0.001
Lab−like – Lab
BH p 0.010
0.0
0.1
0.2
0.3
Pairwise PERMANOVA R2

C


Wild – Wild−like
33/234 (14.1%)
Wild – Lab−like
116/234 (49.6%)
Wild – Lab
131/234 (56%)
Wild−like – Lab−like
144/234 (61.5%)
Wild−like – Lab
156/234 (66.7%)
Lab−like – Lab
87/234 (37.2%)
0
50
100
150
200
FDR−significant predicted KEGG pathways

D


Higher toward lab−associated
Higher toward wild−associated
Wild – Wild−like
Wild – Lab−like
Wild – Lab
Wild−like – Lab−like
Wild−like – Lab
Lab−like – Lab
Metabolism
Biosynthesis of other secondary metabolites (n = 25)
Xenobiotics biodegradation and metabolism (n = 21)
Glycan biosynthesis and metabolism (n = 18)
Metabolism of terpenoids and polyketides (n = 17)
Lipid metabolism (n = 16)
Carbohydrate metabolism (n = 15)
Amino acid metabolism (n = 14)
Metabolism of cofactors and vitamins (n = 12)
Energy metabolism (n = 8)
Metabolism of other amino acids (n = 7)
Nucleotide metabolism (n = 2)
Genetic information processing
Replication and repair (n = 6)
Folding, sorting and degradation (n = 5)
Translation (n = 4)
Transcription (n = 3)
Information processing in viruses (n = 2)
Environmental information processing
Signal transduction (n = 17)
Membrane transport (n = 3)
Signaling molecules and interaction (n = 1)
Cellular processes
Cell growth and death (n = 8)
Transport and catabolism (n = 7)
Cell motility (n = 4)
Cellular community − prokaryotes (n = 4)
Cellular community − eukaryotes (n = 1)
Human diseases
Infectious disease: bacterial (n = 11)
Drug resistance: antimicrobial (n = 3)
100
50
0
50
100
FDR−significant pathways within KEGG Level 2 category (%)